JEE–MAIN EXAMINATION – APRIL 2026

(HELD ON SATURDAY 04th APRIL 2026) **TIME : 3:00 PM TO 6:00 PM**

CHEMISTRY | TEST PAPER WITH SOLUTION

SECTION-A

51. The correct order of total number of atoms of present in

(A) 2 moles of cyclohexane

(B) 684 g of sucrose

(C) 90.8 L of dihydrogen at STP

(1) C > A > B (2) C > B > A

(3) B > C > A (4) B > A C

Sol. (A) Mole of atoms in C_6H_{12}

$= 2 \times 18$

$= 36$

(B) Moles of atoms of sucrose $= \frac{604}{342} \times 45$

(C) $n_{H_2} = \frac{90.8}{22.7} = 4$ moles

Mole fo atoms $= 4 \times 2 = 8$

(B) > (A) > (C)

52. The species having identical radii according to the Bohr's theory are :

A. H (first orbit)

B. He^+ (first orbit)

C. He^+ (second orbit)

D. Li^{2+} (first orbit)

E. Be^{3+} (Second orbit)

Choose the correct answer from the options given below :

(1) A and C Only (2) A and E Only

(3) B and E Only (4) C and D Only

Sol. (A) → same

(B) → same

(C) → same

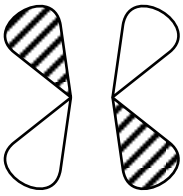
(D) → $r = \frac{90 \times (1)^2}{3} = \frac{90}{3}$

(E) → same

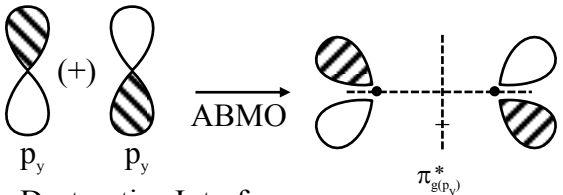
53. Which of the following pictorial diagram most correctly represents the π* (π – antibonding) molecular orbital between two atoms if the internuclear axis is taken to be in the z-direction ($\xrightarrow{z-axis}$) ?

(1) 

(2) 

(3) 

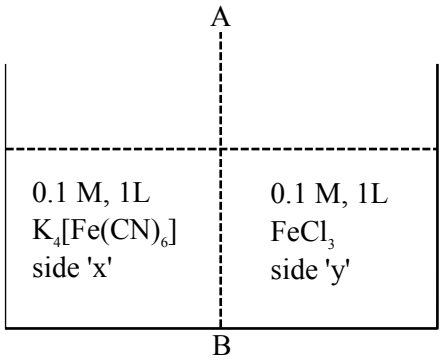
(4) 

Sol. 

Destructive Interference

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54. At 27° C, 0.1 M, 1 L $K_4[Fe(CN)_6]$ aqueous solution and 0.1 M, 1 L $FeCl_3$ aqueous solution are placed in a container separated by a semi permeable membrane AB. Assume complete dissociation of both the solutes. Which of the following statement is ***correct*** ?



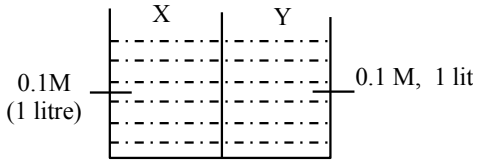
(1) Blue color is formed on both sides.

(2) Ionic solutes in aqueous solution can pass through semi-permeable membrane.

(3) Solution on side 'y' is hypotonic.

(4) To cause the reverse flow of solvent during osmosis, external pressure (any value) should be applied to side 'x'.

Sol.



55. 20 mL of a solution of acetic acid required 28.4 mL of 0.1 M NaOH for its neutralization. A solution (X) was prepared by mixing 20 mL of the above acetic acid and 14.2 mL of 0.1 M NaOH solution. What is the pH of the solution (X) ? (pK_a value of acetic acid is 4.75).

(1) 7.0 (2) 4.45

(3) 3.5 (4) 4.82

Sol. In experiment 1, 28.4 ml of NaOH is required for complete neutralisation, therefore for 14.2 ml, half equivalence point will be achieved to form acidic buffer.

56. Match the **LIST-I** with **LIST-II**

List-I Reaction		**List-II Mechanism**	
A.	Williamson Synthesis	I.	Electrophilic addition
B.	Friedel Craft Reaction	II.	Free radical substitution
C.	Bromination of vinyl benzene	III.	Nucleophilic substitution
D.	Chlorination of toluene in light	IV.	Electrophilic substitution

Choose the ***correct*** answer from the options given below:

(1) A-III, B-I, C-II, D-IV

(2) A-III, B-IV, C-II, D-I

(3) A-III, B-IV, C-I, D-II

(4) A-I, B-III, C-IV, D-II

Sol. Theory based

57. The 1st ionization enthalpy for Mg is +737 kJ/mol. The most probable estimated value of the 2nd ionization enthalpy of Mg is_____.

(1) – 906 kJ/mol

(2) –856 kJ/mol

(3) +1450 kJ/mol

(4) +590 kJ/mol

Sol. Since successive ionization energy increases thus IE_2 (Mg) > IE_1 (Mg)

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58. The electronegativity of a group 13 element 'E' is same as that of Ge (on Pauling scale and upto one decimal point). The **CORRECT** statements about E^{3+} are

A. It can act as a reducing agent.

B. It can act as an oxidizing agent.

C. E^{3+} is more stable than E^{+}.

D. The standard electrode potential value for E^{3+}/E is positive.

Choose the correct answer from the options given below

(1) A and C Only (2) B and C Only

(3) B and D Only (4) A and D Only

Sol. EN of 'Tℓ' is 1.8 that is same as 'Ge'

'E' = Tℓ

$T\ell^{+3}$ acts as oxidizing agent

Stability order : $T\ell^{+3} < T\ell^{+1}$ (due to inert pair effect)

The standard electrode potential value $E°(T\ell^{3+}/T\ell)$ = +1.26 V

59. Pairs of elements with the same number of electrons in their respective 4f orbital are

[Atomic number, Eu-63, Gd-64, Dy-66, Ho-67, Tm-69, Yb-70, Lu-71, Hf-72]

A. (Eu and Gd)

B. (Dy and Ho)

C. (Yb and Hf)

D. (Lu and Tm)

Choose the correct answer from the options given below:

(1) B and C Only (2) A and B Only

(3) A and D Only (4) A and C Only

Sol. Gd – $[Xe]4f^7\ 5d^1\ 6s^2$

Eu – $[Xe]4f^7\ 6s^2$

Dy – $[Xe]4f^{10}\ 6s^2$

Ho – $[Xe]4f^{11}\ 6s^2$

Hf – $[Xe]4f^{14}\ 6s^2\ 5d^2$

Yb – $[Xe]4f^{14}\ 6s^2$

Lu – $[Xe]4f^{14}\ 5d^1\ 6s^2$

Tm – $[Xe]4f^{13}\ 6s^2$

60. Consider the metal complexes $[Ni(en)_3]^{2+}$ (A), $[NiCl_4]^{2-}$(B) and $[Ni(NH_3)_6]^{2+}$(C). Choose the **CORRECT** option by considering the number of unpaired electron present in (A), (B) and (C) respectively and the order of frequency of absorption.

(1) 2, 2, 2 and (A) > (C) > (B)

(2) 0, 2, 0 and (A) > (C) > (B)

(3) 2, 2, 0 and (B) > (C) > (A)

(4) 2, 2, 2 and (C) > (A) > (B)

Sol. (A) $Ni^{2+} \Rightarrow 3d^8$, SFL , $t_{2g}^{2,2,2}\ e_g^{1,1} \Rightarrow$ unpaired $e^- = 2$

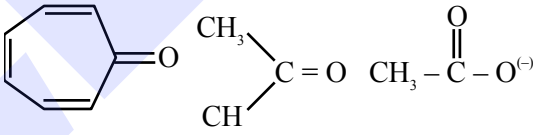
(B) $Ni^{2+} \Rightarrow 3d^8$, WFL , tetrahedral, $e^{2,2}\ t_2^{2,1,1}$ $\Rightarrow$ unpaired $e^- = 2$

(C) $Ni^{2+} \Rightarrow 3d^8$, SFL , $t_{2g}^{2,2,2}\ e_g^{1,1} \Rightarrow$ unpaired $e^- = 2$

As $\Delta_o > \Delta_t$ and Ligand strength : en > NH_3 > Cl^-

Strength of ligand increases, Δ increases, hence more is the frequency required of absorption.

61. Consider the following molecules/species:



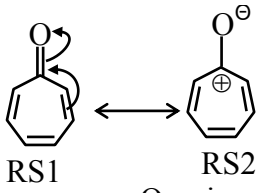
(x) (y) (z)

The correct order of carbon – oxygen double bond length is:

(1) x > y > z (2) y > z > x

(3) z > x > y (4) x > z > y

Sol.



RS1 RS2

Quasi aromatic

C–O bond length is maximum in compound (b) because RS2 of this compound has aromatic character.

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62. Consider |x| is the difference in oxidation states of Mn in highest manganese fluoride and highest manganese oxide. The ions with |x| number of unpaired electrons from the following are:

A. Sc^{3+} B. Zn^{2+}

C. V^{2+} D. Fe^{2+}

E. Co^{2+}

Choose the correct answer from the options given below:

(1) A and B Only (2) C, D and E Only

(3) C and E Only (4) B and E Only

Sol. Oxide of Mn in its highest O.S. = Mn_2O_7 [Mn^{+7}]

Fluoride of Mn in its highest O.S. = MnF_4 [Mn^{+4}]

$\Rightarrow$ Difference in O.S. = |x| = 3

$Sc^{+3} \Rightarrow$ [Ar] ; 0 unpaired e^-

$Zn^{2+} \Rightarrow$ [Ar] $3d^{10}$; 0 unpaired electrons

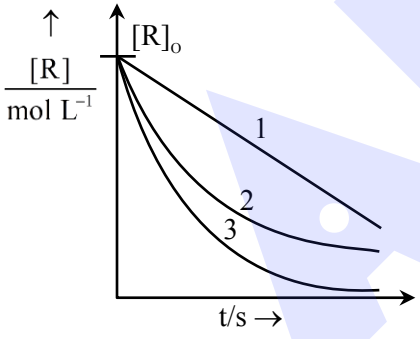
$Fe^{+2} \Rightarrow$ [Ar] $3d^6$; 4 unpaired electrons

$Co^{+2} \Rightarrow$ [Ar] $3d^7$; 3 unpaired electrons

$V^{+2} \Rightarrow$ [Ar] $3d^3$; 3 unpaired electrons

63. Consider the given graph showing variation of reactant concentration with time.

Three different reactions were started with identical initial concentration of reactants. Which of the following statement is correct ?



(1) The order of all the three reactions is same.

(2) The rate constant of reaction 3 is larger than the rate constant of reaction 2 if the order of reaction is same for both.

(3) The SI unit of rate constant of reaction 1 is s^{-1}.

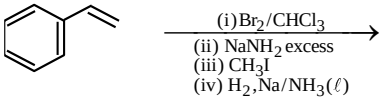
(4) Thermal decomposition of HI on gold surface is an example of reaction 2.

Sol. $\Rightarrow$ Curve Ist represent zero order kinetics

$$[R_t] = [R]_0 - Kt$$

$\Rightarrow K_3 > K_2$

$\Rightarrow$ Reaction-1 is zero order having unit of rate constant same as rate of reaction i.e., molarity / second.

64. 

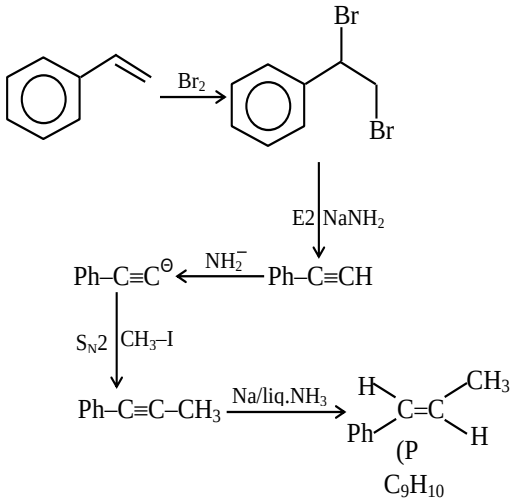
Compound (X) is subjected to the sequence of reactions as shown above. Molar mass of the major product (Y) formed is _____ $gmol^{-1}$.

(Given molar mass in g mol^{-1} C : 12, H : 1 , O : 16)

(1) 90 (2) 118

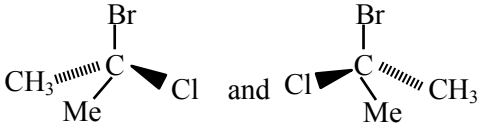
(3) 160 (4) 125

Sol.



Molar mass of Y = 118 gm

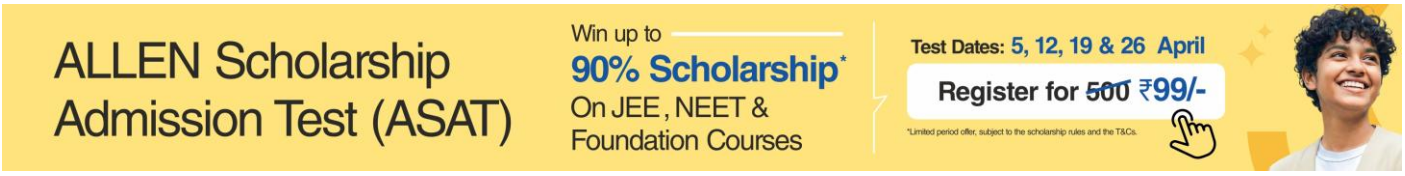
65. The following structures are



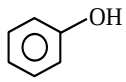
(1) enantiomer (2) Identical molecules

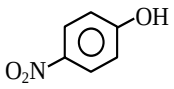
(3) Diastereomers (4) meso compound

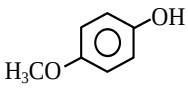
Sol. No chiral center (2 methyl groups)

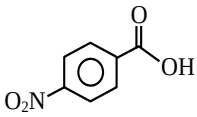


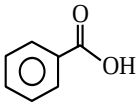
66. The descending order of acidity among the following compounds is

A. 

B. 

C. 

D. 

E. 

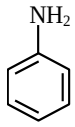
Choose the correct answer from the option given below options

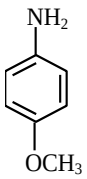
(1) B > D > E > A > C (2) D > B > E > A > C

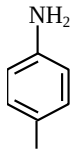
(3) C > A > B > D > E (4) D > E > B > A > C

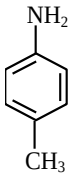
Sol. Theory based.
Benzoic acid is more acidic than phenol.
Electron withdrawing group in increases acidicity.

67. The strongest conjugate acid will result from

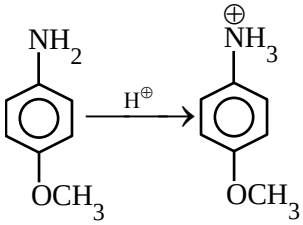
(1) 

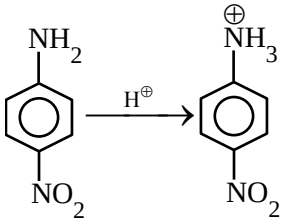
(2) 

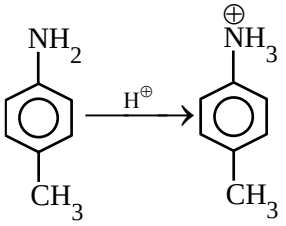
(3) 

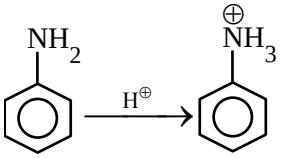
(4) 

Sol.

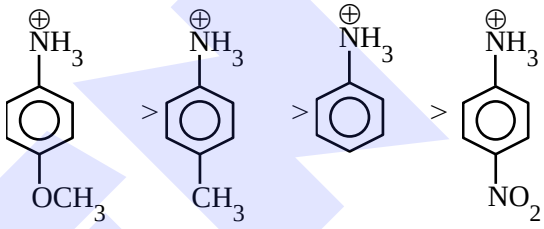






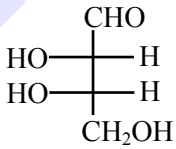


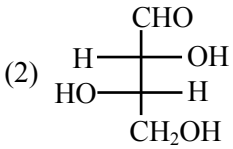
Stability order of conjugated acid :

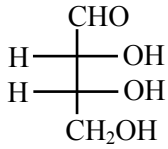


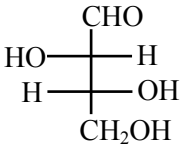
For stronger conjugate acid, we need weaker base.

68. A D-aldotetrose on oxidation with concentrated HNO_3 resulted in optically inactive dicarboxylic acid. The structure of the D-aldotetrose is :

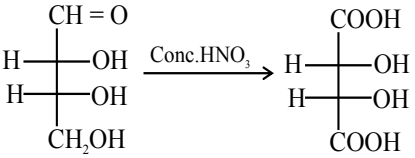
(1) 

(2) 

(3) 

(4) 

Sol.



D-Aldotetrose

Optically inactive Dicarboxylic acid

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69. Among Fe^{3+}, Pb^{2+}, Cu^{2+} and Mn^{2+}, identify the one that gets precipitated out while passing H_2S in presence of NH_4OH as group reagent. The highest possible oxidation state of the corresponding metal is

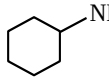
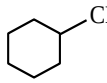
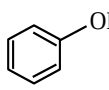
(1) +3 (2) +4

(3) +2 (4) +7

Sol. Group IV reagent : $NH_4OH + H_2S$

Mn^{+2} will be precipitated out and hence highest possible oxidation state of Mn is +7

70. Match the LIST-I with LIST-II

List-I Compound		List-II Test	
A.		I	Hinsberg's reagent test
B.		II	Phthalein dye test
C.		III.	Lucas test
D.		IV.	Tollen's test

Choose the correct answer from the options given below :

(1) A-III, B-I, C-IV , D-II

(2) A-III, B-IV, C-I , D-II

(3) A-I, B-III, C-II , D-IV

(4) A-I, B-II, C-III , D-IV

Sol. Theory based.

Tollen's test is used for test for aldehydes.

Heinsberg's reagent is used to test for amines.

Lucas test is used to test for alcohols.

Phthalein dye test is used for test of phenols.

SECTION-B

71. If 3.365 g of ethanol (ℓ) burnt completely in a bomb calorimeter at 298.15K, the heat produced is 99.472 kJ. The $|\Delta H_f^\circ|$ of ethanol at 298.15 K is _____ $\times 10^2$ kJ mol^{-1}. (Nearest integer)

Given :

Standard enthalpy of combustion of graphite $= -393.5$ kJ mol^{-1}

Standard enthalpy of formation of water (ℓ) $= -285.8$ kJ mol^{-1}

Molar mass in $gmol^{-1}$ of C, H , O are 12, 1 and 16 respectively.

Sol. $\Delta U^\circ_{combustion} = \frac{-q}{moles} = \frac{-99.472}{3.365/46} \Rightarrow 1359.8$ KJ/mole

$C_2H_5OH(\ell) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(\ell)$; $\Delta H_r^\circ = \Delta H_C^\circ$

$\Delta H_C^\circ = \Delta U^\circ + \Delta n_g RT$

$= -1359.8 + \frac{(-1)\times 8.314 \times 298.15}{1000}$

$= -1362.27$ KJ/mole

$-1362.27 = [2 \times (-393.5) + 3 \times (-285.8)] - \Delta H^\circ_{f\ C_2H_5OH(\ell)}$

$-1362.27 = -787 - 857.4 - \Delta H^\circ_{f\ [C_2H_5OH(\ell)]}$

$\Delta H^\circ_{f\ [C_2H_5OH(\ell)]} = -2.82 \times 10^2$ KJ/mole

$\simeq -3 \times 10^2$

72. For the following reaction at 50°C and 2 atm pressure,

$2N_2O_5(g) \rightleftharpoons 2N_2O_4(g) + O_2(g)$

N_2O_5 is 50% dissociated

The magnitude of standard free energy change at this temperature is x.

x = _____ J mol^{-1} [Nearest integer]

Given : R = 8.314 $mol^{-1} K^{-1}$, log 2 = 0.30,

log 3 = 0.48, ln 10 = 2.303 , °C + 273 = K

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Sol. $2N_2O_5\,(q) \rightleftharpoons 2N_2O_4\,(g) + O_2(g)$

t=0	1	–	–
$t_{eqm.}$	1–2x	2x	x
	1–0.5	0.5	0.25
	0.5	0.5	0.25
	$\frac{0.5}{1.25}\times 2$	$\frac{0.5}{1.25}\times 2$	$\frac{0.25}{1.25}\times 2$

$$K_P = \frac{P^2_{N_2O_4}.P_{O_2}}{P^2_{N_2O_5}} = 0.4$$

$\Delta G° = -2.303\ RT \log K_P$

$= -2.303 \times 8.314 \times 323 \log (4/10)$

$= 2473.81$ J/mole

73. An electrochemical cell. consist of the following two redox couples,

$M^{x+}(aq)/\ M(s)\ \ [E^{\ominus}_{red} = +0.15V]$ and

$Fe^{3+}(aq)\ /\ Fe(s)\ [E^{\ominus}_{red} = -0.036V]$

The cell EMF (E_{cell}) is recorded to be 0.2057 V.

If the reaction quotient of the electrochemical reaction is found to be 10^{-2}, then the value of x is _____(Nearest integer)

[Given : M is a p-block metal and $\frac{2.303RT}{F} = 0.059V$]

Sol. $E_{cell} = E^{o}_{cell} - \frac{0.059}{3x}\log 10^{-2}$

$0.2057 = 0.186 - \frac{0.059}{3x}(-2)$

$3x = \frac{0.059\times 2}{0.0197}$

$x = 1.99 \simeq 2$

74. For a first order reaction A $\rightarrow$ B

t/ min	[A]/M
0	0.6500
x	0.0650
20	0.00065

x = _____ min. (Nearest integer)

Sol. $K = \frac{1}{t}\ln\left[\frac{A_0}{A_t}\right]$

$K = \frac{1}{20}\ln\left[\frac{0.6500}{0.00065}\right] \Rightarrow \frac{3\ln 10}{20}\text{min}^{-1}$

Also, $x = \frac{1}{K}\ln\left[\frac{A_0}{A_x}\right] \Rightarrow \frac{20}{3\ln 10}\ln\left(\frac{0.6500}{0.0650}\right)$

$= \frac{20}{3}\text{min} \Rightarrow 6.7\,\text{min}$

75. In sulphur estimation, 2.0×10^{-3} mol of an organic compound (X) (molar mass 76g mol^{-1}) gave 0.4813 g of barium sulphate (molar mass 233 $gmol^{-1}$). The percentage of sulphur in the compound (X) is _____ $\times 10^{-1}$ %

(Nearest integer)

Sol. $\%\ \text{of S} = \frac{\text{Mass of S}}{\text{Mass of BaSO}_4}\times\frac{m}{w}\times 100$

m = gm of $BaSO_4$ ppt

w = gm of Organic compound

$w = 2 \times 10^{-3} \times 76 = 0.152$ gm

$\%\ \text{of S} = \frac{32}{233}\times\frac{0.4813}{0.152}\times 100 = 43.487$

In form of 10^{-1}

$= 43.487 \times 10^{-1} = 434.87 \simeq 435$

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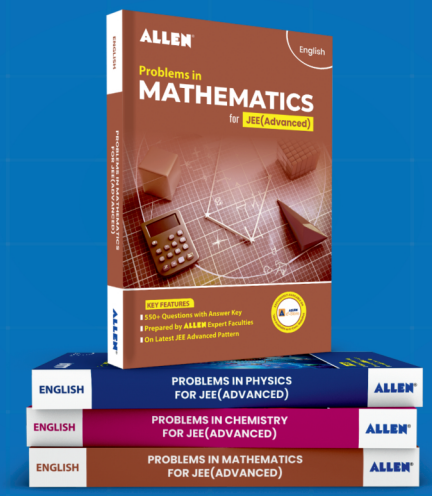


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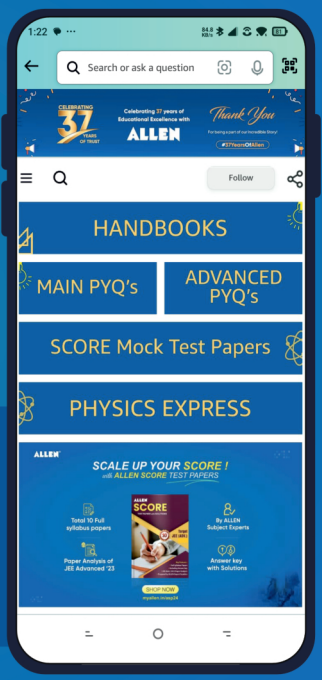
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